

We can express y (target) as a $(k-1)$ dim vector

eg: if $k=4$

$$1 = (1, 0, 0)$$

$$2 = (0, 1, 0)$$

$$3 = (0, 0, 1)$$

$$4 = (0, 0, 0)$$

$$e^{\eta - \ln(1-p)} = \frac{1}{1+e^{-\eta}}$$

$+ \ln\left(\frac{1-e^{-\eta}}{e^{1-k-\eta}}\right)$
 $- \ln(1+e^{-\eta})$

Then if p_i is proba = $p(y=i|\eta)$

$$\Rightarrow p(y) = p_1^{y_1} p_2^{y_2} p_3^{y_3} \dots p_{k-1}^{y_{k-1}} (1 - p_1 - p_2 - \dots - p_{k-1})^{(1-y_1-y_2-\dots-y_{k-1})}$$

$$\therefore \ln(p(y|\eta)) = \exp \left[\sum_{i=1}^{k-1} y_i \ln(p_i) + \ln(1 - p_1 - p_2 - \dots - p_{k-1}) \right]$$

$$= \exp \left[\sum_{i=1}^{k-1} y_i \ln \left(\frac{p_i}{1 - \sum_{i=1}^{k-1} p_i} \right) + \ln \left(1 - \sum_{i=1}^{k-1} p_i \right) \right]$$

$$\text{Let } T(y) = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_{k-1} \end{bmatrix}, \quad \eta = \begin{bmatrix} \ln \left(\frac{p_1}{1 - \sum_{i=1}^{k-1} p_i} \right) \\ \vdots \\ \ln \left(\frac{p_{k-1}}{1 - \sum_{i=1}^{k-1} p_i} \right) \end{bmatrix}$$

$$\Rightarrow \vec{c}^\eta = \begin{bmatrix} \frac{p_1}{1 - \sum p_i} \\ \vdots \\ \frac{p_{k-1}}{1 - \sum p_i} \end{bmatrix} \Rightarrow \frac{\vec{c}^\eta}{1 \cdot e^\eta} = \frac{\sum p_i}{1 - \sum p_i}$$

$$\Rightarrow \frac{1}{1 \cdot e^\eta} + 1 = \frac{1}{\sum p_i} \Rightarrow$$

$$\Rightarrow \sum p_i = \frac{1}{1 + \frac{1}{1 \cdot e^\eta}}$$

$$\therefore 1 - \sum p_i = \frac{\frac{1}{1 \cdot e^\eta}}{1 + \frac{1}{1 \cdot e^\eta}} = \frac{1}{1 \cdot e^\eta + 1}$$

$$\therefore A(\eta) = -\ln(1 - \sum p_i) = \underline{\underline{\ln(1 + \vec{1} \cdot e^\eta)}}$$

GLM assumption : η is linear in features

$$\therefore \eta \text{ shape} = \frac{(k+1) \times 1}{k \times 1}$$

$$x \text{ shape} = n \times 1 \quad (n \text{ features})$$

$$\Rightarrow W = \underline{\underline{n \times k \text{ (b'x'n) matrix}}}$$

$$\vec{\eta} = W\vec{x}$$

probabilities : $p(y=i|\eta) = h(y) \cdot \exp(\eta^T T(y) - A(\eta))$

Since here, $T(y) = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix}$ i^{th} value

$$\therefore p(y=i) = \exp(\eta_i - A(\eta))$$

since $\vec{\eta} = W\vec{x}$, $\eta_i = \underbrace{W_{(i, \cdot)}}_{\text{throw } W} \vec{x}$

$$\therefore p(y=i) = \exp\left(W_{(i, \cdot)} \vec{x} - \ln(1 + \vec{1} \cdot e^\eta)\right)$$

$$= \frac{e^{W_{(i, \cdot)} \vec{x}}}{1 + \vec{1} \cdot e^\eta}$$

↳ This is softmax used in multiclass classification